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Theorem: For any given n, such that $n \in \mathbb{N} \cup \{0\}$, it is verified that:

$$\underbrace{666\ldots6}_{\cancel{n\text{ times}}\ n+1\text{ times}} \times \underbrace{666\ldots67}_{n\text{ times}} = \underbrace{44\ldots4}_{n+1\text{ times}}\underbrace{22\ldots2}_{n+1\text{ times}}$$

Proof: Let $a_n := \sum_{k=0}^{n} 10^k$. We are going to prove the statement above employing the induction principle. Let the following be our induction hypothesis:

$$(6a_n)(6a_n+1) = 2a_n + 4\cdot 10^{n+1} a_n$$

Which is, indeed, equivalent to the aforementioned statement:

- $6a_n = \underbrace{666\ldots6}_{n+1\text{ times}}$; $6a_n+1 = \underbrace{666\ldots67}_{n\text{ times}}$

- $2a_n = \underbrace{222\ldots2}_{n+1\text{ times}}$

- $4a_n 10^{n+1} = \underbrace{444\ldots4}_{n+1\text{ times}}\underbrace{000\ldots0}_{n+1\text{ times}}$

For $n=0$: Our hypothesis is true for $n=0$

$$a_n = \sum_{k=0}^{0} 10^k = 10^0 = 1$$

$$(6\cdot 1+1)\times(6\cdot 1) = 7\cdot 6 = 42 = 2\cdot 1 + 4\cdot 10$$

For $n=n+1$:

$$a_{n+1} = \sum_{k=0}^{n+1} 10^k = \sum_{k=0}^{n} 10^k + 10^{n+1} = a_n + 10^{n+1}$$

$$(6a_{n+1})(6a_{n+1}+1) = 36(a_{n+1})^2 + 6a_{n+1} = 36(a_n+10^{n+1})^2 + 6(a_n+10^{n+1})$$

$$= 36((a_n)^2 + 2a_n 10^{n+1} + 10^{2n+2}) + 6(a_n+10^{n+1})$$

$$= 36(a_n)^2 + 72a_n 10^{n+1} + 36\cdot 10^{2n+2} + 6a_n + 6\cdot 10^{n+1}$$

$$= 2a_n + 2\cdot 10^{n+1} + 4\cdot 10^{n+2} a_n + 4a_n + 4\cdot 10^{n+1} + 32a10^{n+1} + 36a^2$$

$$+ 36\cdot 10^{2n+2} = 2a_n + 2\cdot 10^{n+1} + 4\cdot 10^{n+2} a_n + 4(a_n + 10^{n+1} + 8a_n 10^{n+1}$$

$$+ 9(a_n)^2 + 9\cdot 10^{2n+2})$$

$$= 2a_n + 2\cdot 10^{n+1} + 4\cdot 10^{n+2} a_n + 4(a_n + 10^{n+1} + 8a_n 10^{n+1} + 9(a_n)^2 + \underbrace{10^{2n+3} - 10^{2n+2}})$$

$$= \text{""} + 4((a_n + 10^{n+1} + 8a_n 10^{n+1} + 9(a_n)^2 - 10^{2n+2}) + 10^{2n+3})$$

↳

$$= \text{""} + 4((10a_n + 1 + 8a_n 10^{n+1} + 9(a_n)^2 - 10^{2n+2}) + 10^{2n+3})$$

$$= \text{""} + 4((a_n(10 + 8\cdot 10^{n+1} + 9a_n) + 1 - 10^{2n+2}) + 10^{2n+3})$$

$$= \text{""} + 4((a_n(10 + 9\cdot 10^{n+1} - 1) + 1 - 10^{2n+2}) + 10^{2n+3})$$

$$= \text{""} + 4((a_n(9 + 9\cdot 10^{n+1}) + 1 - 10^{2n+2}) + 10^{2n+3})$$

$$= \text{""} + 4((9a_n(10^{n+1} + 1) + 1 - 10^{2n+2}) + 10^{2n+3})$$

$$= \text{""} + 4(((10^{n+1} - 1)(10^{n+1} + 1) + 1 - 10^{2n+2}) + 10^{2n+3})$$

$$= \text{""} + 4(10^{2n+2} - 1 + 1 - 10^{2n+2} + 10^{2n+3})$$

$$= \text{""} + 4\cdot 10^{2n+3} = 2a_n + 2\cdot 10^{n+1} + 4\cdot 10^{n+2} a_n + 4\cdot 10^{2n+3}$$

$$= 2(a_n + 10^{n+1}) + 4\cdot 10^{n+2}(a_n + 10^{n+1})$$

$$\boxed{= 2a_{n+1} + 4\cdot 10^{n+2} a_{n+1}}$$

And, if we plug in $n+1$ in the second member of our induction hypothesis, we will get this same result, therefore, our hypothesis is also valid for $n = n+1$.

Since our induction hypothesis is valid for $n=0$ & $n=n+1$, ~~and~~ it is therefore valid for all natural numbers and zero, thus, our statement is proven.

Q.E.D.